

Question number	Answer	Additional guidance	Mark
8(a)(i)	An explanation that includes the following points: • (combination of antibiotics given) because different bacteria are resistant to different antibiotics (1) • (given for several months) to expose bacteria to high enough doses for long enough (1)	ACCEPT to ensure all bacteria destroyed	(2)

Question number	Answer	Additional guidance	Mark
8(a)(ii)	An answer that includes three of the following points: • because <i>Mycobacterium</i> infects the lungs (1) • (<i>Mycobacterium</i> infection) destroying lung tissue (1) • therefore reduced <u>gas exchange</u> (1) • so insufficient oxygen to meet (oxygen) demands of patient (1)	ACCEPT a description e.g. necrosis IGNORE breathing problems	(3)

Question number	Answer	Additional guidance	Mark
8(a)(iii)	An explanation that includes four of the following points: • because if DNA synthesis cannot take place neither can <u>mitosis</u> (1) • and therefore clonal expansion of {T cells / B cells} cannot take place (1) • without T killer cells, host-infected cells cannot be destroyed (1) • without {B cells / plasma cells} there will be no antibody for opsonisation (of bacteria) (1) • therefore macrophages cannot destroy the <i>Mycobacteria</i> (1)	ACCEPT {T cells / B cells} cannot divide DO NOT ACCEPT B cells release antibody	(4)

Question number	Answer	Additional guidance
)	Indicative content: • number of bacteria still increasing at start of phage therapy (D) • because it takes time for lytic cycle (E) • bacteria number decrease (D) • because cells get destroyed when viruses burst out (E) • due to lytic cycle (E) • virus numbers increase (D) • because viruses replicate inside bacteria (E) • and burst out (E) • virus numbers increase after bacterial numbers increase (D) • because it takes time for synthesis of new components (E) • virus particles decrease (D) • because they get {engulfed / destroyed} by macrophages (E) • cyclical increase and decrease in the number of bacteria (D) • because not all bacteria get infected (E) • therefore surviving bacteria reproduce and increase in number (E) • cyclical increase and decrease in number of viruses (D) • because macrophages cannot destroy all viruses (E) • because viruses infect more bacteria (E)	Level 1 : 1 mark = 1 comment made about the graph or viral infections 2 marks = 3 comments made about {graph / viral infections} OR 1 description and a linked explanation Level 2 : 3 marks = 2 descriptions and linked explanations 4 marks = 3 descriptions and linked explanations Level 3 : 5 marks = 4 descriptions and linked explanations that includes both virus and bacterial number changes 6 marks = 5 descriptions and linked explanations that includes both virus and bacterial number changes + the cyclical nature of the data

	• eventually no viruses or bacteria (D) • because viruses destroy bacteria faster than they can reproduce (E) • viruses have no more host cells / destroyed by macrophages (E)		
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